

Dependence of the Bremsstrahlung Spectral Temperature in Minimum-B Electron Cyclotron Resonance Ion Sources

J. Benitez, C. Lyneis, L. Phair, D. Todd, and D. Xie

Abstract—The spectral temperature T_s obtained from bremsstrahlung spectra emitted from electron cyclotron resonance (ECR) ion sources (ECRISs), in which the plasma is confined in a minimum-B magnetic field, is used as a relative indication of the temperature of the plasma hot electrons. Past bremsstrahlung measurements taken on ECRISs indicate that T_s is strongly dependent on the magnetic field gradient at the resonance zone or $(B_{\min}/B_{\text{ECR}})$. However, this dependence was never fully proven or explained. To further our understanding a more detailed study of the bremsstrahlung radiation for X-rays above 10 keV is underway using VENUS, a third-generation ECRIS at Lawrence Berkeley National Laboratory. Initial analysis of previous and new data has revealed that T_s appears to be dependent solely on the minimum magnetic field $B_{\min}$ rather than $(B_{\min}/B_{\text{ECR}})$ and the microwave frequency ω . Decoupling T_s from $(B_{\min}/B_{\text{ECR}})$, mainly B_{ECR} , implies that T_s does not depend on the magnitude of the heating frequency ω . While it certainly appears to be true that plasma density $n_e (\propto \omega^2 \propto B^2)$ increases with heating frequency, as was postulated by Geller in 1987, a more careful consideration into the heating mechanism of the plasma electrons is warranted. The disassociation of T_s from the heating frequency ω , while an interesting discovery, implies that we must change the way we understand how ECRISs operate. This paper presents new bremsstrahlung measurements, analyses, and discussions of the results.

Index Terms—Bremsstrahlung, electron cyclotron resonance (ECR) plasma, plasma hot electrons, X-ray detectors.

I. INTRODUCTION

IN THE mid-1960s and 1970s, Geller introduced electron cyclotron resonance heating as means to heat plasma electrons and produce multiply charged ions. In the 1980s Geller [1] showed that with MINIMAFIOS-16 GHz, higher extraction current and charge state distributions can be achieved with higher microwave frequencies. A consequence of using higher microwave frequencies is the required higher magnetic field. The performance of high-charge state electron cyclotron resonance ion sources (ECRISs) scales roughly with the square of microwave heating frequency, as predicted by

Geller [1]. The best present ECRIS performers are the third generation ECRISs operating at either 24 or 28 GHz [2]–[5] with accordingly higher magnetic fields than earlier generation sources operating at frequencies of 18 GHz or lower. These third generation ECRISs also produce significantly more intense and more energetic bremsstrahlung.

VENUS is a third generation ECRIS which uses both 18- and 28-GHz microwave powers to produce a plasma. The electrons in a VENUS plasma can gain sufficient energy to produce high-charge states such as Ar^{18+} [6], Xe^{43+} , and U^{50+} [2]. The ionization potentials for these highly charged ions are in the range of 2–5 keV and the maximum electron impact ionization rate occurs at electrons of energy of three to five times the ionization potentials, i.e., approximately 6 to 25 keV. Bremsstrahlung spectra recorded from VENUS shows that this superconducting ECR produces plasma with electrons of maximum energy up to several hundred keV.

Because VENUS is a superconducting ECRIS, the high energy bremsstrahlung is an important consideration since it adds a thermal load to the cryogenic system. The plasma chamber and outer wall of the cryostat are transparent to the X-rays above ~ 300 keV and their energy can be deposited into the cold mass [7]. If not properly addressed, this bremsstrahlung could severely deteriorate the performance of the ECRIS' superconducting magnet system. This paper presents an investigation into how the various parameters of the magnetic field configuration can change the hot tail of bremsstrahlung emitted from VENUS and the spectral temperatures T_s . The results are particularly important for the development of fourth generation ECRIS that will operate above 40 GHz [8], and are likely to produce more intense and energetic bremsstrahlung.

II. EXPERIMENTAL SETUP

VENUS uses three superconducting solenoids and one superconducting sextupole to generate the minimum-B magnetic field configuration to confine the plasma, as shown in Fig. 1. Each coil is independently powered and the axial fields can produce up to 4 T at injection, 3 T at extraction, and a center coil that can produce a $B_{\min}$ value from 0 to 2 T. Fig. 1 shows an example of the shape of the axial magnetic field in VENUS. The sextupole coils can produce up to 2.2 T on the plasma wall at a radius of 72 mm [9] and the radial field generated by the sextupole is typically set to values close to the peak axial field at extraction. This produces a closed resonance

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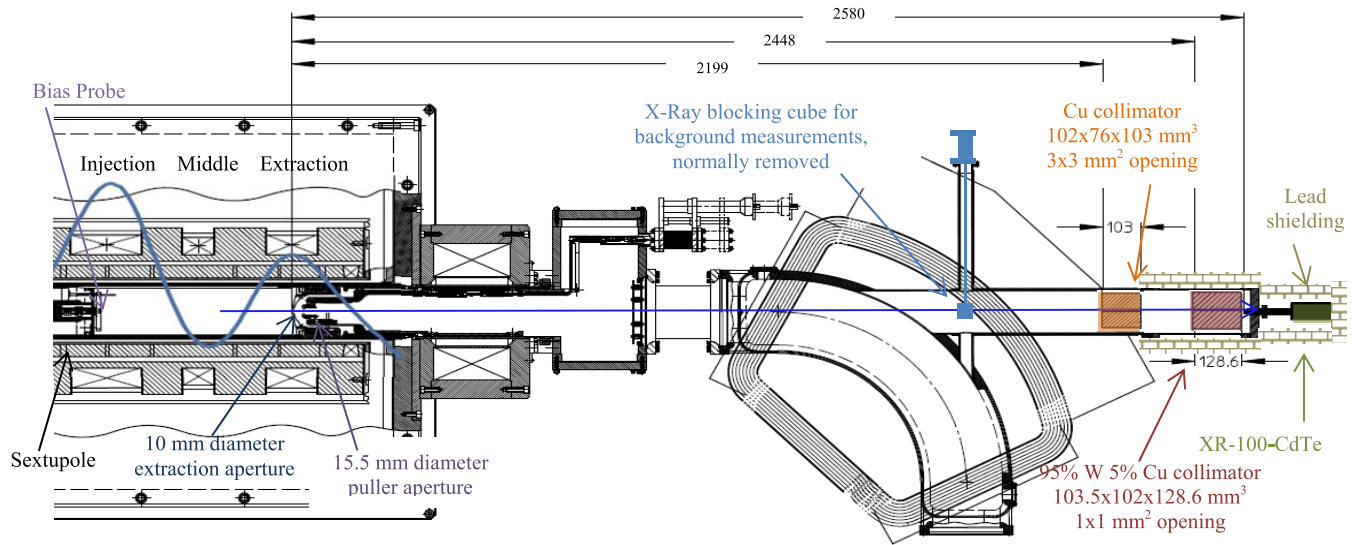


Fig. 1. Schematic of the experimental setup of the XR-100-CdTe detector at the VENUS ion source. All units in mm.

field at 2 T or twice B_{ECR} at 28 GHz. A key difference between VENUS and the previous generation ECRISs is the presence of a center coil, in addition to the injection and extraction coils. This adds the flexibility of easily varying the field gradient ∇B_{ECR} and the center field B_{min} either together or with one held approximately fixed. The central field can be lowered sufficiently so there are simultaneous resonance fields for 14, 18, and 28 GHz. The VENUS microwave system consists of a 10-kW 28-GHz gyrotron and a 2-kW 18-GHz klystron. A 2-kW 14-GHz klystron was also used briefly to test the frequency dependence of the spectral temperature.

The axially emitted bremsstrahlung spectra from VENUS were recorded using an Amptek XR-100-CdTe X-ray and γ -ray diode detector [10]. It consists of a 1-mm thick, 25-mm² detector that sits behind a 0.1-mm Be window. Its resolution at 122 keV is <1.5 keV Full Width Half Maxima, which is not as good as Si diode detectors. However, the CdTe detectors are much more sensitive at energies above 20 keV as they have a higher stopping power with shorter attenuation lengths [11]. In addition, the XR-100-CdTe detector has the advantage that it can be operated in vacuum, without attenuation caused by windows. The CdTe detector energy calibration was done using standard radioactive sources with known gamma lines.

From 4 to 80 keV the intrinsic efficiency, a ratio of the number of pulses recorded by the detector to the number of gamma rays hitting the detector, with the window taken into account, is above 80%. However, this paper includes X-rays measured between 10 and 300 keV. Between 80 and 300 keV the efficiency drops gradually to 10%. All spectra shown have been corrected for detector intrinsic efficiency which includes the correction for the 4-mil Be window on the detector. The detector's intrinsic efficiency was verified using various radioactive sources with known activities and compared well to that quoted by the manufacturer and this was used to make the corrections.

The detector sits on the optical axis, in line with the extraction electrode center, and inside the vacuum chamber,

as shown in Fig. 1. Therefore, no corrections due to attenuation in air or through a window are necessary. The spectra are not corrected for the detector's photo-peak efficiency, the probability of producing a full-energy peak.

Between the detector and the aluminum extraction aperture is a set of two collimators. The first sits directly up against the detector face and is a heavy metal (95% W and 5% Cu) block. It is 128.6 mm long and has a 1-mm square-groove. The second collimator is 146-mm upstream of the first collimator and is solid Cu. It is 102 mm long with a 3-mm square-groove. This setup corresponds to a square with 20 mm sides at the extraction electrode.

With the collimation in place, the bremsstrahlung reaching the detector comes from three different regions close to the axis of VENUS. First, electrons which strike the bias probe at the injection end of VENUS produce bremsstrahlung which may reach the detector, however, most are forward facing in direction. Second, bremsstrahlung produced in the volume visible to the detector in collisions between the electrons and ions in the plasma can reach the detector. Third, electron losses at the extraction region generate bremsstrahlung when they strike the plasma extraction electrode within an area defined by the acceptance of the collimator. The bremsstrahlung generated at extraction must have sufficient energy to pass through the aluminum plasma electrode.

It is important to note that midway through the measurements, which spanned several months, the 10 mm diameter extraction aperture was replaced with a slightly thicker aluminum electrode at the aperture opening to improve heat transfer and prevent melting it. Analysis of spectra recorded before and after show that this only affected their shapes at the low-energy end, which is not discussed in this paper. Although the new electrode decreases the total counts by up to 50%, the slope of the spectra in the energy range between 80 and 300 keV did not change, as shown in Fig. 2. This evidence suggests that for the bremsstrahlung between 80 and 300 keV, the dominant source of detected counts were the electrons striking the plasma electrode. At low

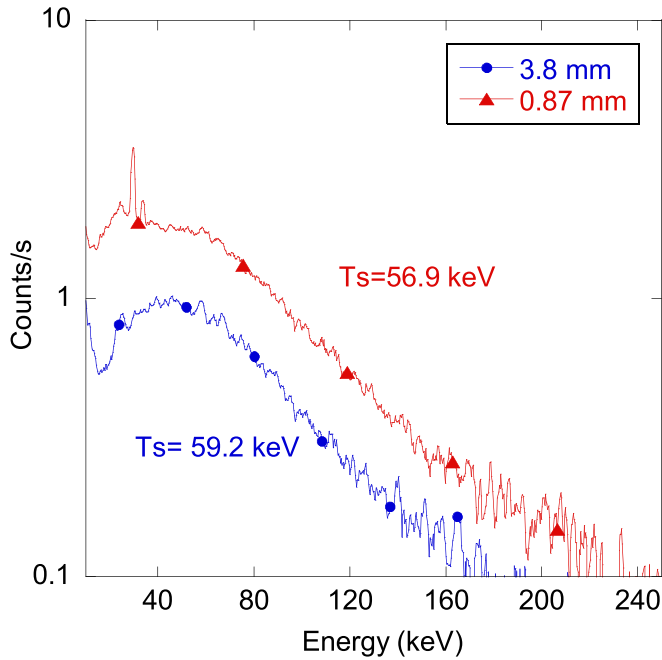


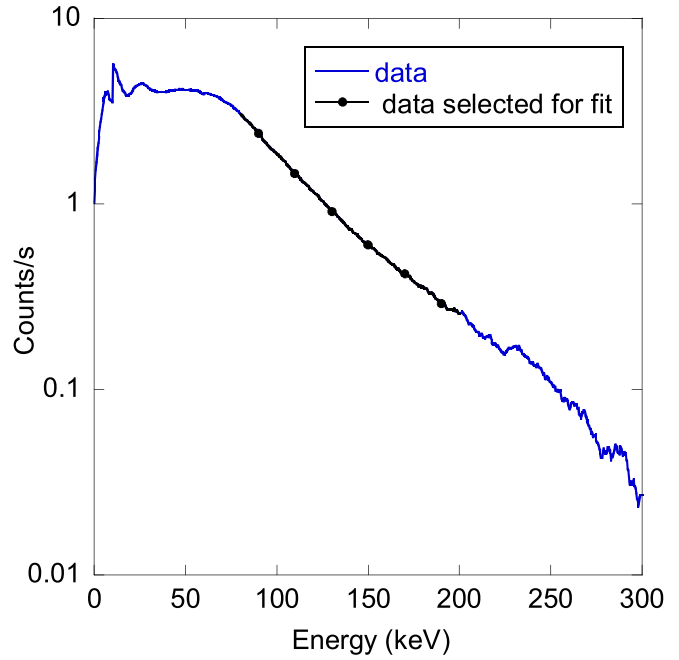
Fig. 2. Changing the thickness of the aluminum extraction electrode reduced the total number of counts but did not change the shape of the spectra or the spectral temperature.

energies the plasma electrode serves as an additional collimator.

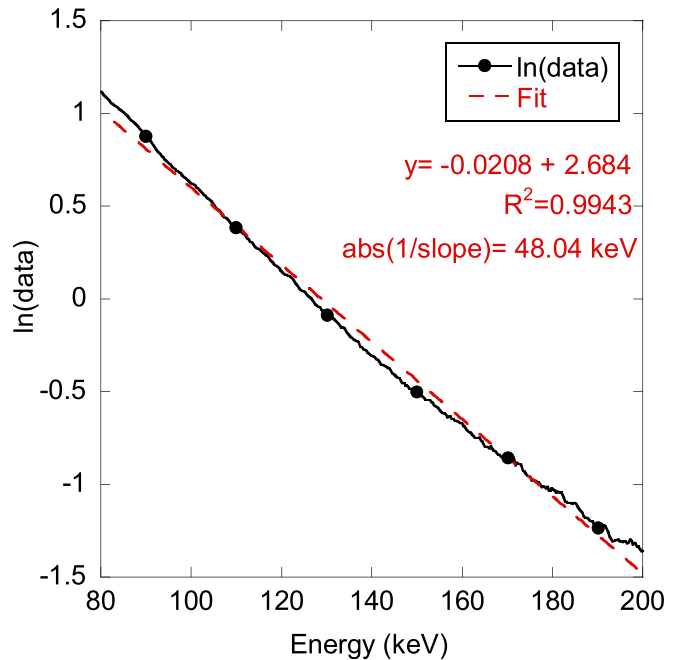
III. SPECTRAL TEMPERATURE

Investigation of plasma bremsstrahlung radiation is often done by using a spectral temperature T_s which is used in this paper and elsewhere as a relative indication of electron population temperature. In the high-energy range of the bremsstrahlung spectra one can assume that the spectral power $j(\hbar\omega)$ follows a Maxwell-Boltzmann distribution: $j(\hbar\omega) \propto \exp(-\hbar\omega/T_s)$ [12]. When a semi-log plot of the bremsstrahlung spectrum $j(\hbar\omega)$ versus energy, such as shown in Fig. 3(a), is made over a range of photon energies, it can be fit with a straight line. The reciprocal of the line's slope yields a spectral temperature T_s in keV. In Fig. 3(a), the energy range 80–200 keV was selected and the natural log of this data range is plotted in Fig. 3(b). This spectral temperature is a relative indication of the temperature of the hot electrons, not a direct measure of the temperature distribution of the hot electrons in the plasma. Extraction of the final electron energy distribution would require a complex deconvolution of the measured spectra with corrections for conversion from electron impact to bremsstrahlung as well as corrections for collimation and detector efficiencies and this is beyond the scope of this paper.

An X-ray blocking cube, shown in Fig. 1, was installed to measure the background reaching the detector due to less than complete collimation. When the 22 mm on a side cube made from a tantalum-niobium alloy is inserted, it blocks the X-rays from entering the collimators and reaching the detector. The remaining X-rays detected are from indirectly scattered bremsstrahlung and can be subtracted as a back-



a.



b.

Fig. 3. (a) Data for determining T_s is typically taken in the energy range 80–200 keV. (b) Inverse of the slope of a linear fit of the semi-log plot of the data yields a spectral temperature.

ground. A background spectrum is shown in Fig. 4(b) and is compared to an original spectrum, both not yet corrected for efficiency. The background spectrum shows peaks at 58, 59.3, and 67.2 keV, which are the tungsten characteristic X-ray lines $K\alpha_2$, $K\alpha_1$, and $K\beta_1$ lines, respectively. The background spectrum below 100 keV is negligible but it does contribute a large percentage of the counts at higher energies and this affects the spectral temperature extracted from the

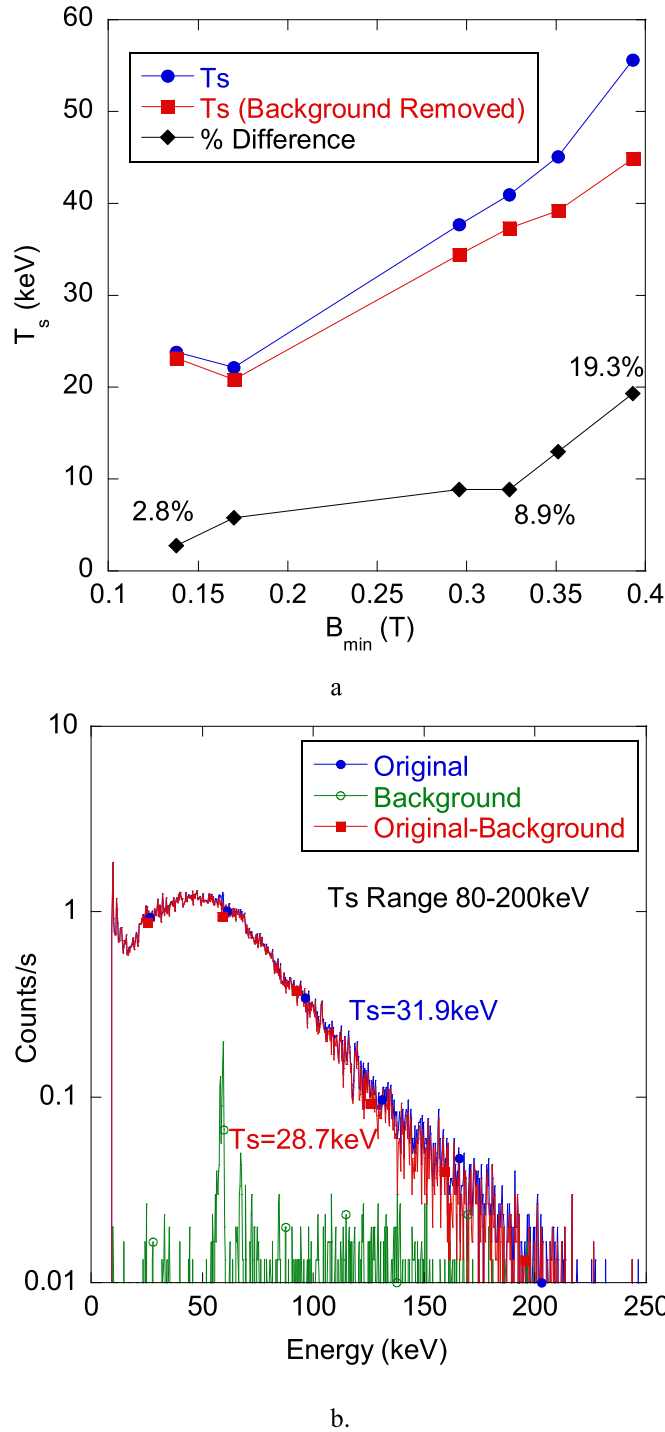


Fig. 4. (a) Spectra is shown with and without the background subtracted. (b) Removing the background increases the discrepancy in T_s as $B_{\min}$ is increased, since the total counts increase with $B_{\min}$.

data. When the background is subtracted from the original spectra, the spectral temperature is reduced by 2% to 20%. Fig. 4(a) and (b) shows that the background becomes more significant when $B_{\min}$ is increased since this increases the relative number of higher energy electrons in the spectrum as well as the amount of scattered X-rays. All data presented are spectra without background subtraction and this adds a systematic error, up to 20% or higher, to the results. However, the focus of the investigation is on the dependence of T_s on a

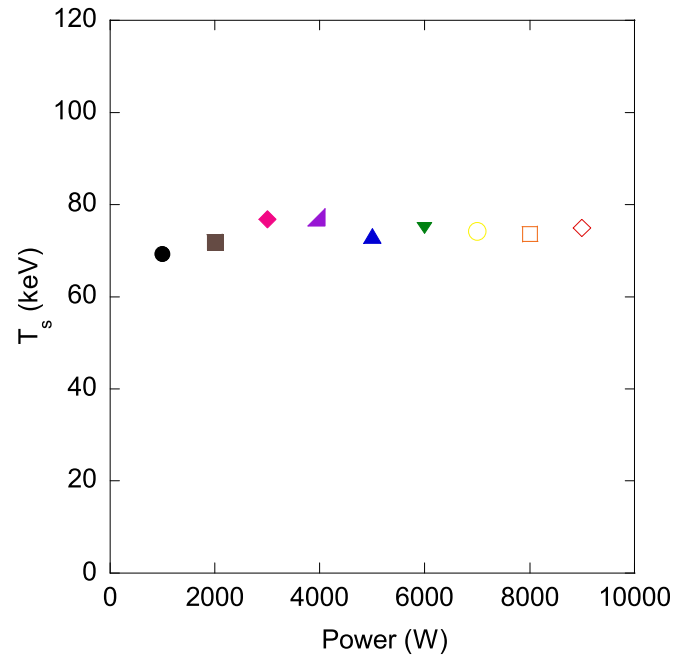


Fig. 5. Measured T_s shows essentially no changes with increasing microwave power from 1000 to 9000 W with VENUS at a fixed magnetic field configuration. The spectra were recorded with combined 18- and 28-GHz heating frequencies.

few important parameters, not the absolute magnitude of the spectral temperature.

IV. EXPERIMENTAL RESULTS

A. Experimental Parameters

Aside from the magnetic field configuration and heating frequency the measurements were made with similar parameters through all data sets. The microwave power level essentially does not change the T_s in ECR plasma, though the bremsstrahlung radiation rate is proportional to the power level [13]. Shown in Fig. 5 is the measured T_s with VENUS at a fixed magnetic field configuration for microwave power up to 9000 W, indicating just relatively small variations. To avoid an unstable plasma generating large fluctuations in the bremsstrahlung, all data were recorded at a power level of 1000 W, measured at the transmitter, for 14-, 18-, or 28- GHz heating frequencies, and at similar oxygen gas reference pressure of $\sim 2 \times 10^{-7}$ torr at the injection region and bias probe voltage.

We focused on how the magnetic configuration and microwave heating frequencies affect the axial bremsstrahlung and the resulting spectral temperature. The investigation was divided into three main sections. First, $B_{\min}$ was varied while the gradient at the resonance zone B_{ECR} was held approximately constant. Second, the gradient at the resonance zone B_{ECR} was varied while $B_{\min}$ was held constant. Third, we also compared spectra recorded at a fixed magnetic field configuration for three different heating frequencies. The gradient at the resonance zone B_{ECR} , henceforth written as ∇B_{ECR} , is calculated on axis.

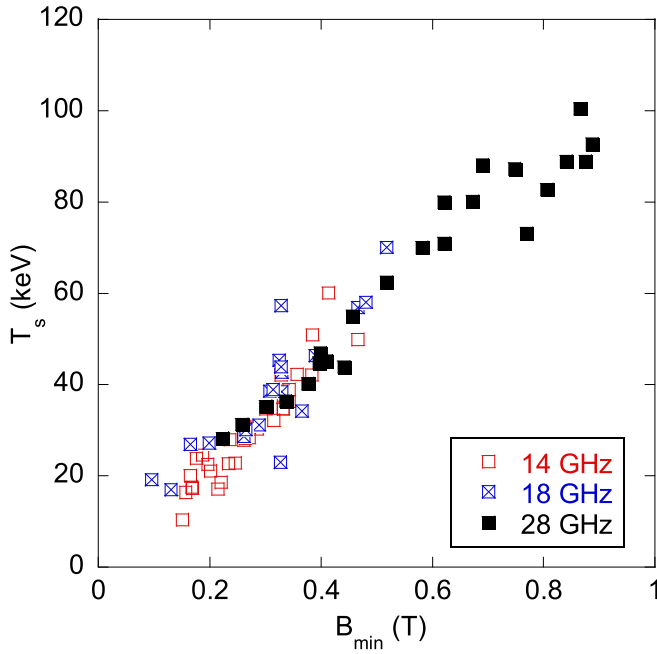


Fig. 6. All T_s measured with VENUS in this new study is plotted versus $B_{\min}$.

TABLE I
RESULTS FOR CONSTANT GRADIENT AT B_{ECR}

f (GHz)	$B_{\min}$ on axis (T)	Inj. ∇B_{ECR} (T/m)	Ext. ∇B_{ECR} (T/m)	ECR Volume (cm^3)	T_s (keV)
18	0.26	-7.5	6.77	1655.1	30.2
	0.31	-8.16	6.44	1319.8	38.9
	0.33	-7.64	7.03	1327.8	38.5
	0.36	-7.54	6.44	1235.3	46.6
	0.38	-7.45	6.32	1138.1	55.0
	0.48	-7.71	6.54	322.7	58.0
28	0.26	-12.66	10.98		31.3
	0.30	-12.44	10.75	1990.9	35.2
	0.34	-12.29	10.62	1923.1	36.4
	0.38	-12.05	10.09	1849.4	40.2
	0.41	-11.15	9.88	1786.4	45.1
	0.62	-11.13	8.74	1174.6	80.0

Magnetic field settings with approximately constant gradient at the resonance zone and variable $B_{\min}$ for 18 and 28 GHz heating.

Shown in Fig. 6 is a graphic summary of the measurements in which the T_s is plotted versus the minimum center field $B_{\min}$ for three different independently used heating frequencies of 14, 18, and 28 GHz, at various field configurations and gradients. This summary plot clearly shows that the hot electron spectral temperature T_s depends linearly on $B_{\min}$ only and does not depend on the microwave heating frequency ∇B_{ECR} or the mirror ratios for the VENUS operating fields.

B. Constant Gradient at B_{ECR} While Varying $B_{\min}$

In the first part of the investigation ∇B_{ECR} was held approximately constant but $B_{\min}$ was varied for the three heating frequencies 28, 18, and 14 GHz. For the case of 18-GHz heating, ∇B_{ECR} was varied by 9% while $B_{\min}$ was varied by 59%. The parameters used for 18-GHz heating are listed in Table I. Also listed are the field parameters for the

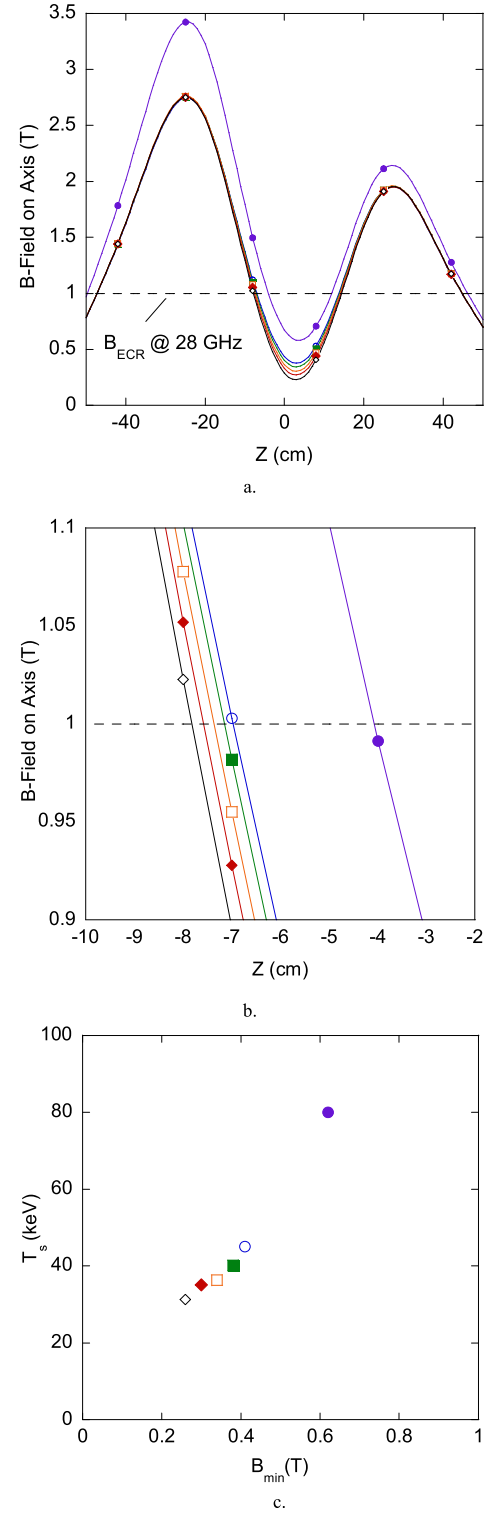


Fig. 7. Using 28-GHz heating frequency: (a) and (b) axial magnetic fields on axis for quasi-constant gradient at the resonance zone and (c) spectral temperatures.

case of 28-GHz heating, in which ∇B_{ECR} was varied by 13% while $B_{\min}$ was varied by 82%. It is seen when ∇B_{ECR} is held approximately constant and $B_{\min}$ is varied this changes T_s significantly. The experiment was also done with 14-GHz heating frequency and similar results were seen. Fig. 7 shows the magnetic field configuration and T_s versus $B_{\min}$ for the 28-GHz settings.

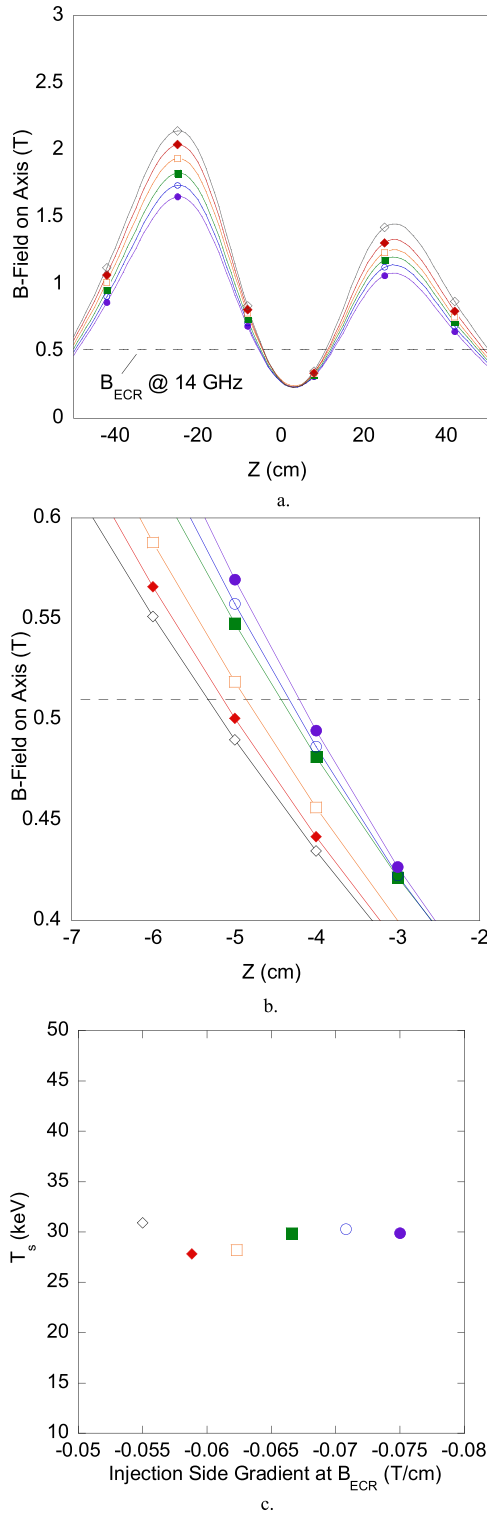


Fig. 8. (a) and (b) Axial magnetic fields on axis for constant $B_{\min}$ and (c) spectral temperatures for the case of 14-GHz heating.

C. Constant $B_{\min}$ While Varying Gradient at B_{ECR}

In the second part of the investigation we held $B_{\min}$ constant and varied ∇B_{ECR} for 14 and 18 GHz. Fig. 8(a) and (b) illustrates how ∇B_{ECR} was varied while maintaining a constant $B_{\min}$ for the 14-GHz tests. With the minimum field and radial field held constant at 0.26 and 1 T, the injection and extraction

TABLE II
RESULTS FOR CONSTANT $B_{\min}$

f (GHz)	$B_{\min}$ on axis (T)	Inj. ∇B_{ECR} (T/m)	Ext. ∇B_{ECR} (T/m)	ECR Volume (cm ³)	T_s (keV)
14	0.26	-6.13	5.09	1448.3	30.9
	0.26	-5.94	5.40	1416.9	27.8
	0.26	-6.30	5.40	1355.5	28.2
	0.26	-6.69	5.66	1276.6	29.9
	0.26	-7.15	6.14	1272	30.3
	0.26	-7.58	6.26	1239.5	29.9
18	0.33	-7.24	6.16	1472	42.8
	0.33	-7.64	7.03	1327.8	43.8
	0.33	-8.91	7.65	1227.7	38.4
	0.33	-9.25	8.41	1148.6	45.3

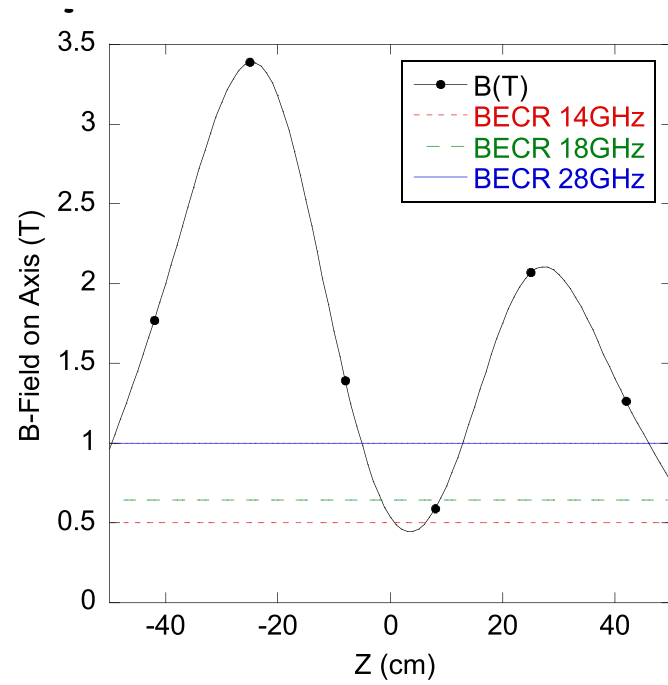
Magnetic field settings with approximately constant $B_{\min}$ and variable gradient at the resonance zone for 14 and 18 GHz heating.

fields were varied from 1.64 to 2.14 T and 1.07 to 1.45 T. This corresponds to a change of mirror ratio of 6.3–8.2 and 4.1–5.6 at injection and extraction, respectively, which results in a substantial variation of ∇B_{ECR} .

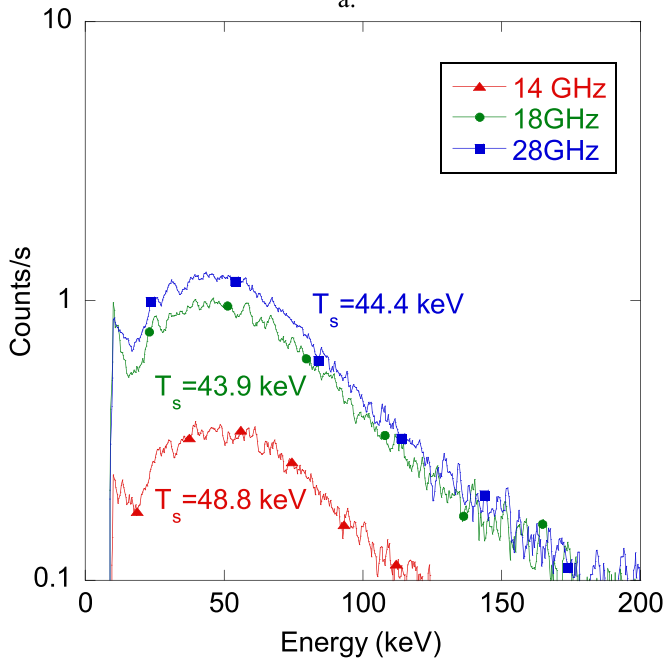
Column 6 of Table II shows that the small variations of the spectral temperature, within the measurement error, are not directly correlated with the gradient that varies up to about 28% at injection and 23% at extraction regions for the case of 14-GHz heating. Fig. 8(c) shows the resulting T_s versus $B_{\min}$ for the 14-GHz settings. For 18 GHz, the minimum field and radial field were held constant at 0.33 and 1.29 T. The injection and extraction fields were varied from 1.67 to 2.90 T and 1.14 to 1.93 T corresponding to a change of mirror ratio of 5.1 to 8.8 at the injection and 3.5 to 5.8 at the extraction, respectively. The spectral temperatures obtained using 18-GHz heating do not vary by more than 17% while the gradient at the injection was varied by 36%. The experiment was also done with 28-GHz heating frequency and similar results were seen.

D. Comparison of Different Frequencies With a Fixed Magnetic Field Configuration

Spectra were recorded at the same magnetic field configuration, shown in Fig. 9(a), with each heating frequency 14, 18, and 28 GHz. Fig. 9(b) shows spectra for a fixed magnetic field configuration shown in Fig. 9(a). The shape of the spectra does not depend on the heating frequency used, although the count rate differs. For all spectra the microwave power was set to 1 kW at the exit of the klystron or gyrotron. There was a mismatch in between the 14-GHz klystron and VENUS plasma chamber resulting in about 10% reflected power which may explain the lower count rate. However, the tests show that the spectral temperature was insensitive to the differences in these microwave power levels. While the magnetic field configuration was kept constant, for different heating frequencies this results in significantly different gradients at each resonance zone as shown in Table III. For example in this test ∇B_{ECR} at injection changes from -4.67 T/m at 14 GHz to -12.75 T/m and the spectral temperature remains approximately constant. This supports the idea that the spectral temperature is not correlated with the magnetic field gradient



a.



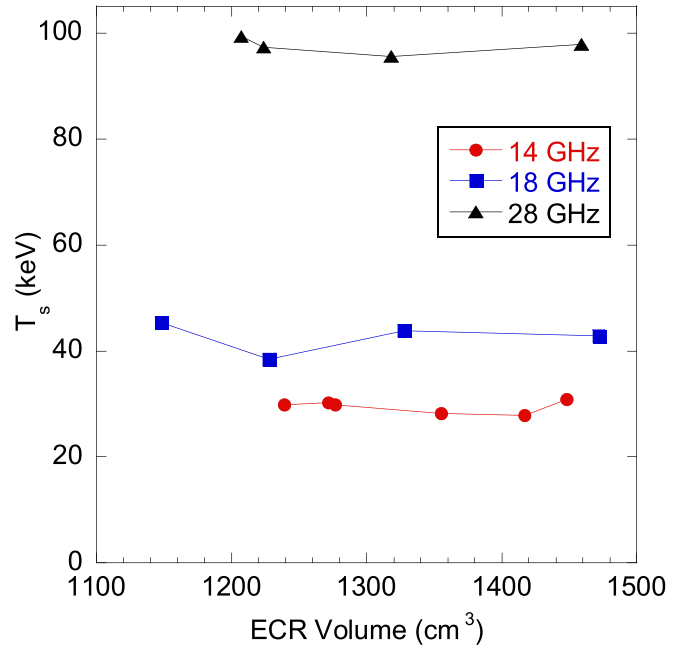
b.

Fig. 9. (a) Axial magnetic field on axis used for spectra recorded at 14-, 18-, and 28-GHz heating frequencies. (b) Resulting spectra and T_s .

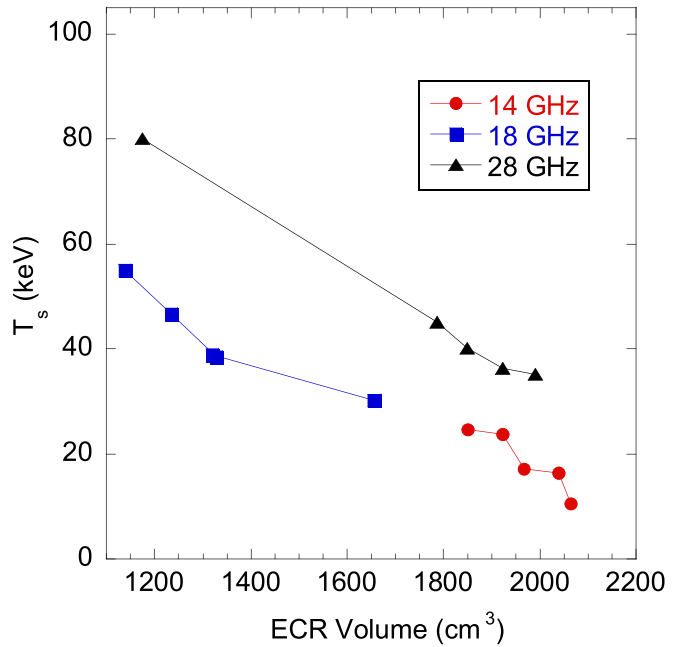
at resonance. Although only one set of measurements is presented, it was shown true for several configurations.

E. Effect of ECR Volume

The dependence of T_s on the ECR volume is also investigated. Fig. 10(a) shows the spectral temperature T_s versus the ECR volume in which $B_{\min}$ is held constant and T_s is insensitive to the ECR volume. However, Fig. 10(b) shows



a.



b.

Fig. 10. Dependence of T_s on ECR volume. (a) When only $B_{\min}$ is held constant the results show T_s is insensitive to the ECR volume. (b) When only the gradient at the resonance zone is held constant T_s changes with volume because $B_{\min}$ is also changing. For each data set B_{rad} is held constant.

that if only the gradient at the resonance zone is held constant T_s does change with ECR volume since $B_{\min}$ is also changing. These results indicate that neither the gradient at the resonance zone nor the ECR volume has substantial effects on the spectral temperature T_s . These results are examined further in Section V. Note that at each data set for each frequency the radial field is held constant.

TABLE III
RESULTS FOR CONSTANT MAGNETIC FIELD CONFIGURATION

f (GHz)	Inj. ∇B_{ECR} (T/m)	Ext. ∇B_{ECR} (T/m)	T_s (keV)
14	-4.67	4.22	48.8
18	-7.71	6.54	43.8
28	-12.75	10.00	44.4

Results for spectra recorded at identical magnetic field configuration with each heating frequency. The injection, extraction, and middle axial fields were 3.39T, 2.10T, and 0.48T.

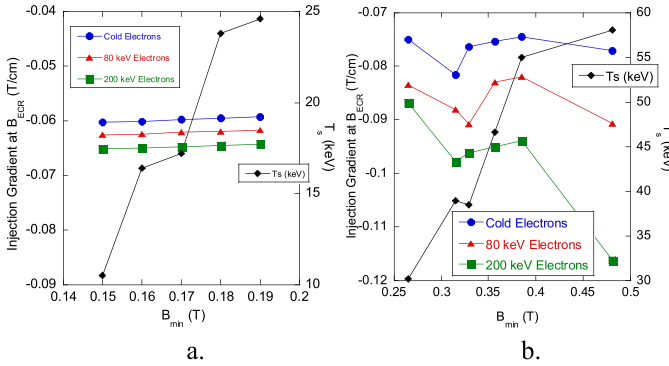


Fig. 11. Dependence of the gradient at resonance for different energy electrons. (a) Gradients at the resonance zone of 80- and 200-keV electrons with 14-GHz heating frequency. (b) Gradients at the resonance zone of 80- and 200-keV electrons with 18-GHz heating frequency.

F. Relativistic Considerations

As previously mentioned, the spectral temperature T_s is typically calculated from a linear fit in the photon energy range 80–200 keV. However, the field gradients are calculated at the resonance field for the cold electrons. For a given heating frequency, there is a resonant heating surface for cold electrons. This heating surface is surrounded by resonant heating surfaces that correspond to higher energy electrons. One consideration is whether taking into account the gradients for the relativistic electrons will affect the interpretation of the results.

During the study, when varying $B_{\min}$, the axial magnetic field gradient was only looked at for the resonant surface of the cold electrons. However, in Fig. 11, the gradients at the resonance zone for 80 and 200 keV electrons were examined for the settings used to maintain a constant gradient at the B_{ECR} for cold electrons. Fig. 11(a) shows that for the settings at 14 GHz the gradients at the resonance zone of 80 and 200-keV electrons remained constant with increasing $B_{\min}$, as it did so for the cold electrons. When the settings used for 28 GHz were examined it was also found that the 80- and 200-keV electrons gradients did not change with increasing $B_{\min}$, similar to that shown in Fig. 11(a). Fig. 11(b) shows that for the settings at 18 GHz the gradients at the resonance zone of 200-keV electrons increased with increasing $B_{\min}$. Historically, it has been expected that spectral temperature should rise with decreased gradient, but here the spectral temperature rises although the cold electron resonance gradient is held constant and the warm electron resonance gradient is increasing. Further, as shown in previous sections, when

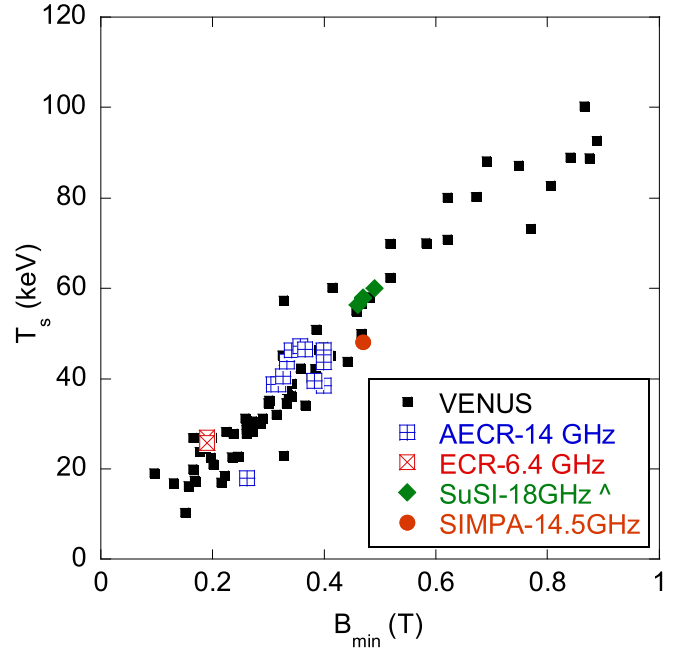


Fig. 12. All T_s measured with VENUS and a number of published data from other ECRISs are plotted versus $B_{\min}$ [12], [16]. It clearly and very convincingly demonstrates the dependence of T_s on $B_{\min}$.

$B_{\min}$ is held constant and the axial resonance zone gradients for warm and cold electrons are allowed to vary, the spectral temperature is constant. These results point to the key parameter involved in determining the spectral temperature is $B_{\min}$.

V. DISCUSSION

In a number of previous studies, there appeared to be a correlation between ∇B_{ECR} and T_s , and this was thought to be evidence that at lower ∇B_{ECR} electron heating is more efficient. However, in these earlier studies both ∇B_{ECR} and $B_{\min}$ were changed at the same time [14], [15]. In addition to the gradient at the resonance zone, it was also believed that the electron energy gained depended on the plasma ECR volume. The results presented indicate that the ECR volume has a little influence on the T_s .

It has often been observed that as an ECRIS is operated at higher frequencies the spectral temperature increases. Furthermore, it was typically thought that it was the ratio of $B_{\min}/B_{\text{ECR}}$ that was the important factor in determining T_s . These results show that T_s depends on not the ratio of $B_{\min}$ to B_{ECR} nor the ECR plasma volume, but simply on the magnitude of $B_{\min}$. This means that as new, higher frequency sources come into operation the generation of hot electrons can be controlled if $B_{\min}$ is not scaled up with frequency. In the case of the VENUS operation with $B_{\min}$ at about 0.4 T can produce excellent results without generating large amounts of high energy bremsstrahlung which can penetrate into the cold mass and generate an unwanted cryogenic heat load. Additionally, for fixed magnetic field geometry T_s is independent of whether the microwave frequency is 14, 18, or 28 GHz. This result strengthens the case that T_s is not correlated with ∇B_{ECR} , since with a fixed magnetic geometry, the value of ∇B_{ECR} is different for each frequency.

Plotted versus $B_{\min}$ in Fig. 12 are the T_s measured with VENUS in the new study and a few published measurements from other high-charge state ECRISs operating at substantially different heating frequencies and field configurations [12], [16]. It clearly shows a linear dependence of T_s on $B_{\min}$ which further demonstrates that T_s is mainly a function of $B_{\min}$. The mechanisms behind the correlation of $B_{\min}$ to T_s are not yet clear and are under study.

In summary, this study carried out in a high-charge state ECRIS, VENUS, for the first time clearly demonstrates, that the axial T_s depends linearly on the center field $B_{\min}$. This T_s is insensitive to the magnetic field gradient ∇B_{ECR} , the heating frequency f , the ECR volume, and the magnetic mirrors.

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Authors' photographs and biographies not available at the time of publication.